PES University - Dept. of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #02: Campus & Academic Operations

Automated Rubric Assignment Evaluator

Functional Requirements (FR-001 to FR-005)

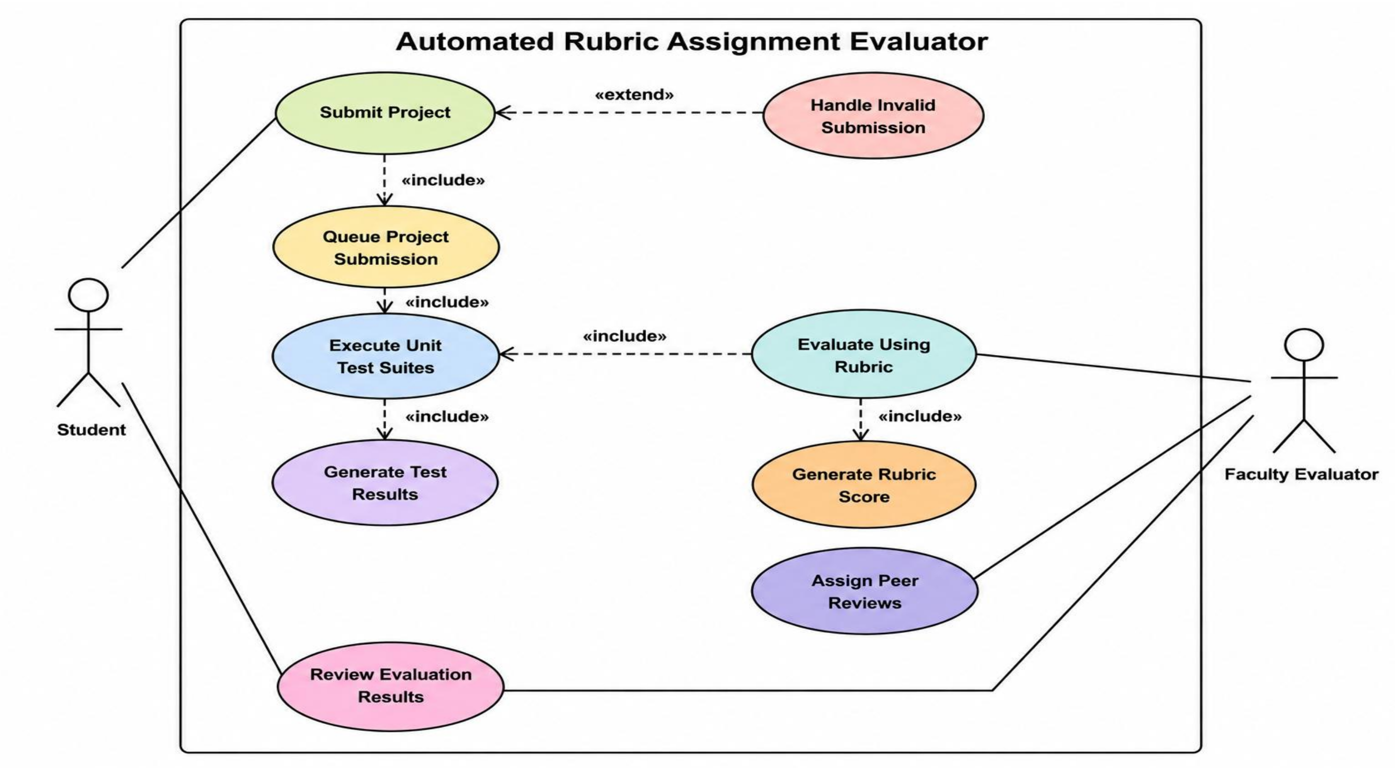
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional — Submission & Evaluation	The system shall automatically queue uploaded student project zip files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	**Pass:** Rubric breakdown is rendered with test results within 10 seconds of submission. **Fail:** File queue stalls or invalid rubric tally.	Automates the core evaluation bottleneck, removing manual test execution and scoring overhead for faculty.
FR-002	Functional — Peer Review Allocation	The system shall automatically assign each submission to a configurable number of peer reviewers, ensuring double-blind anonymity (reviewer and reviewee identities hidden from each other) and balanced distribution across the class.	High	**Pass:** Every submission receives the configured number of reviewers with no reviewer assigned their own submission, and identity fields are masked in the review UI. **Fail:** Unassigned submissions remain after allocation, or any identity leak is detected.	Eliminates manual peer-review distribution overhead, which is explicitly called out as a pain point in the problem context.
FR-003	Functional — Score Override & Audit	The system shall allow a Faculty Evaluator to manually adjust an auto-generated rubric score for any rubric item, with a mandatory justification comment, and shall log the change in an audit trail.	Medium	**Pass:** Overridden score, original score, evaluator ID, timestamp, and comment are all persisted and retrievable. **Fail:** Override is accepted without a justification comment, or audit record is missing/incomplete.	Automated scoring cannot capture all evaluation nuance (e.g., plagiarism, partial credit for logic); faculty need a controlled, traceable override path.
FR-004	Functional — Student Feedback & Resubmission	The system shall notify students of their rubric results and test outcomes immediately after evaluation, and shall allow resubmission of the project within a configurable deadline window.	Medium	**Pass:** Notification (in-app/email) is triggered within 1 minute of score finalization, and resubmission is accepted only while the window is open. **Fail:** Notification is not sent, or resubmission is accepted after the deadline window closes.	Timely feedback closes the loop for students and supports iterative learning within a bounded academic timeline.
FR-005	Functional — Analytics & Reporting	The system shall generate a class-wide analytics dashboard for Faculty Evaluators, summarizing average rubric-item scores, submission timeliness, and	Low	**Pass:** Dashboard reflects current submission data and updates within 5 minutes of new evaluations . **Fail:** Dashboard shows stale data older than the refresh	Gives faculty aggregate visibility into class performance and common failure patterns without

		flagged (failed) test cases across all students.		threshold or omits any evaluated submission.	manually compiling results.

Non-Functional Requirements (NFR-001 & NFR-002)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding 1.5 GB memory footprint.	High	**Pass:** Benchmarking tests confirm target latency and security standards under simulated peak load . **Fail:** Queue drops items or memory footprint exceeds 1.5 GB under load.	Submission spikes near deadlines are common in academic settings; the system must remain stable and secure under this predictable peak load.
NFR-002	Reliability & Availability	The system shall maintain at least 99.5% uptime during active submission and review windows (as configured per assignment deadline), with automatic recovery/restart on failure within 60 seconds.	High	**Pass:** Uptime monitoring logs confirm ≥99.5% availability during the configured window, and simulated crash recovery completes within 60 seconds. **Fail:** Downtime exceeds the 0.5% threshold during an active window, or recovery takes longer than 60 seconds.	Since submissions and peer reviews are tightly bound to hard deadlines, downtime directly and unfairly penalizes students; high availability during these windows is critical to fairness.

UML Use-Case Diagram:



Use-Case Flow Specification

Use Case: Upload Project Submission

Use-Case ID: UC-01 Primary Actor: Student Secondary Actors: System (Test Runner, Rubric Engine), Faculty Evaluator (receives output) Related Use Cases:

«include» Execute Test Suite

«include» Assign Peer Reviews

«extend» Request Resubmission (extends this use case when resubmission is permitted)

Brief Description

A Student uploads a zipped project submission before the assignment deadline. The system queues the file, automatically triggers the test suite execution and rubric scoring workflow, and assigns peer reviewers — without any manual intervention from the Faculty Evaluator.

Preconditions

PRE-1: The Student is authenticated and enrolled in the course/assignment.

PRE-2: The assignment submission window is currently open (current time ≤ deadline).

PRE-3: A rubric configuration and unit test suite have already been set up by the Faculty Evaluator for this assignment.

PRE-4: The Student has a valid project archive (.zip) ready on their local device.

Postconditions

POST-1 (Success): The submission is stored, queued, evaluated, an itemized rubric score breakdown is generated, and peer reviewers are assigned to the submission.

POST-2 (Success): The Student can view their rubric results, and the submission becomes visible to assigned peer reviewers.

POST-3 (Failure): No partial or corrupted submission record is persisted; the Student is informed of the failure reason and prompted to retry.

Main Success Scenario

The Student navigates to the assignment page and selects Upload Submission.

The Student selects a project

.zip

file from their device and confirms the upload.

The system validates the file (format, size limit, non-empty archive).

The system stores the file and adds it to the processing queue with a timestamp.

The system triggers the Execute Test Suite use case

«include»

, running the pre-configured unit tests

against the submission.

The system generates an itemized rubric score breakdown based on the test results and static rubric criteria.

The system triggers the Assign Peer Reviews use case

«include»

, allocating the submission to peer

reviewers under double-blind rules.

The system displays a confirmation message to the Student, including the submission timestamp and queue position.

The Student later views the rubric score breakdown once evaluation completes (within 10 seconds per NFR target).

Use case ends successfully.

Alternate Flow A1: Invalid or Corrupted File

Triggered at step 3 of the Main Success Scenario.

The system detects that the uploaded file is not a valid

.zip

, exceeds the size limit, or is empty/corrupted.

The system rejects the upload and displays a specific error message (e.g., "Invalid file format" or "File exceeds 50 MB limit").

The system does not add the file to the processing queue.

The Student is prompted to select a new file and re-attempt the upload.

The flow returns to step 2 of the Main Success Scenario.

(If the deadline has since passed during retries, the extending use case

Request Resubmission

governs whether a

late upload is still permitted.)

Alternate Flow A2: Submission Deadline Passed

Triggered at step 1, before file selection.

The Student attempts to access the upload interface after the deadline has elapsed.

The system checks whether late submission / resubmission is enabled for this assignment (per Faculty Evaluator configuration).

If disabled, the system blocks the upload and displays "Submission window closed."

If enabled, control passes to the Request Resubmission extension point, which governs late-window

eligibility and any penalty rules before resuming at step 2 of the Main Success Scenario.

Special Requirements

Response time for steps 3–4 (validation and queuing) must not exceed 2 seconds under normal load.

The system must support concurrent uploads from up to 100 students without dropping queue items (per NFR-001).

Frequency of Use

High — this is the primary entry point into the system and is expected to be triggered by every enrolled student, once per submission attempt, near each assignment deadline